

1 Mining Complex Data Types

A. Sequence Data

1. Time-Series Data
 - Equal time intervals
 - Stock prices
 - Temperature logs
 - Similarity Search
 - DFT
 - DWT
 - Trend Analysis
 - Long-term Trends
 - Cyclic Movements
 - Seasonal Variations
 - Random Movements
2. Symbolic Sequences
 - Event-based data
 - Clickstreams
 - Shopping history
3. Biological Sequences
 - DNA
 - Proteins
 - BLAST
 - Hidden Markov Models (HMM)

B. Graphs & Networks

1. Homogeneous Networks
 - Same node/link types
 - Facebook friendship network
2. Heterogeneous Networks
 - Different node/link types
 - Healthcare network
3. Key Concepts
 - Power Law Distribution
 - Small World Phenomenon
 - Densification
 - Shrinking Diameter

C. Other Data Modalities

1. Spatial Data
2. Spatiotemporal Data
3. Cyber-Physical Systems (CPS)
4. Multimedia Data
5. Text Mining
6. Web Mining
 - Content Mining
 - Structure Mining
 - Usage Mining
7. Data Streams
 - Sliding Window Algorithms

2 Advanced Methodologies & Foundations

A. Statistical Data Mining

1. Regression
2. Generalized Linear Models (GLM)
3. ANOVA
4. Factor Analysis
5. Survival Analysis

B. Theoretical Foundations

1. Data Reduction Theory
2. Data Compression Theory
 - MDL Principle
3. Microeconomic View
4. Pattern Discovery / Inductive Databases

3 Real-World Applications

A. Financial Data Analysis

1. Loan Prediction
2. Credit Profiling
3. Money Laundering Detection

B. Retail & Telecommunication

1. Targeted Marketing
2. Customer Retention
3. Loyalty Programs
4. Fraud Detection

C. Intrusion Detection Systems (IDS)

1. Signature-Based Detection
 - Known attacks
 - Cannot detect zero-day attacks
2. Anomaly-Based Detection
 - Detects new attacks
 - High False Positives

4 Recommender Systems

A. Content-Based Filtering

1. Uses item features
2. User profile matching
3. Similar item recommendation

B. Collaborative Filtering (CF)

1. Memory-Based CF
 - k-NN
 - Pearson Correlation
 - Cosine Similarity
2. Model-Based CF
 - Bayesian Networks
 - Clustering
 - Machine Learning Models

C. Evaluation

1. False Negatives
2. False Positives
3. Hybrid / Ensemble Systems
 - Netflix Prize Concept

5 Privacy-Preserving Data Mining (PPDM)

A. Randomization

- Add noise to data

B. k-Anonymity

1. Generalization
2. Suppression

C. l-Diversity

- Diverse sensitive values

D. Distributed Privacy Preservation

1. Horizontal Partitioning
2. Vertical Partitioning
3. Cryptographic Protocols

E. Differential Privacy

- Individual record protection

6 Future Trends of Data Mining

1. Scalable Mining
2. Interactive / Constraint-Based Mining
3. Cloud-Based Mining
4. Search Engine Integration
5. Real-Time Analytics
6. Stream Data Mining

DATA MINING: FUTURE, APPLICATIONS & SOCIETY

1. INTRODUCTION TO WEB MINING

Definition

- Application of Data Mining techniques on World Wide Web.
- Goal:
Discover, extract and analyze useful knowledge automatically.

Why Web Mining?

- Huge amount of web data exists.
- Manual analysis impossible.
- Helps search engines, e-commerce, recommendation systems and social media.

Data Mining vs Web Mining

Data Mining	Web Mining
• Structured data. • Relational databases. • Rows, columns, keys. • Example: - Student database.	• Semi-structured data. • Unstructured data. • HTML pages, text, images, audio and videos. • Example: Google Search.

2. TYPES OF WEB DATA

Web Content

- Actual content inside webpage.
- Includes:
Text
Images
Videos
Audio
Records



Web Structure

- Relationships among pages.
- Hyperlinks.
- HTML/XML tags.

Intra-Page Structure

- Structure within a page.
- HTML tags.

Inter-Page Structure

- Hyperlinks between pages.

Web Usage

- User browsing behavior.
- Clickstreams.
- HTTP Logs.
- Server Logs.



Supplemental Data

- Cookies.
- User profiles.
- Registration information.
- Session information.

WEB MINING

3. WEB CHARACTERISTICS & CHALLENGES

- **Huge Volume**
• Massive amount of information.
- **Wide Coverage**
• Almost every topic available.
- **Semi-Structured Nature**
• Nested HTML documents.
- **Highly Linked**
• Pages connected by hyperlinks.
- **Redundancy**
• Same information appears on multiple websites.
- **Noise**
• Advertisements.
• Menus.
• Copyright notices.
- **Dynamic Nature**
• New pages continuously added.
• Existing pages continuously updated.



4. THREE MAIN CATEGORIES OF WEB MINING

WEB MINING

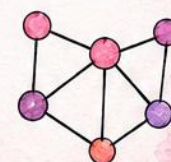
Web Content Mining



Web Usage Mining



Web Structure Mining



A. WEB CONTENT MINING

Purpose

- Extract useful information from webpage contents.

Data Sources

- Text
- Images
- Videos
- Audio
- Graphs



Goal

- Find most relevant results for user queries.

Techniques

Classification

- Assign documents to predefined classes.

Example:

Science Articles
Political Articles
Sports Articles

Clustering

- Group similar documents.
- No predefined classes.

Example:

Similar research papers
grouped automatically.

Summarization

- Produce shorter version of large documents.
- Preserve key information.



Visualization

- Graphical representation.

data web
mining text
text-info



Examples:

Word Clouds
Heat Maps
Scatter Plots
Tree Maps

Applications

- Search Engines
- News Aggregators
- Digital Libraries



B. WEB USAGE MINING

Definition

- Analysis of user browsing behavior using web logs.

Main Data Source

- Server Log Files
- Browser Logs
- Clickstreams



Goal

- Understand customer behavior.
- Predict future actions.
- Improve website performance.



Three Phases

1. Data Preprocessing

- Data Cleaning
- Data Integration
- Data Transformation
- Session Identification

2. Pattern Discovery

- Apply mining algorithms.
- Discover hidden patterns.

3. Pattern Analysis

- Evaluate discovered patterns.
- Predict user behavior.

Techniques Used

Classification

- Predict class labels.

Examples:

Online Buyer
Non-Buyer

Algorithms:

Decision Trees
Logistic Regression
Neural Networks
KNN



Clustering

- Group similar users.

Goal:

High intra-cluster similarity
Low inter-cluster similarity

Algorithm:

K-Means



Association Rules

- Discover items frequently occurring together.

Example:

Laptop → Mouse
Bread → Butter



Applications

- Customer Analysis
- Website Optimization
- Personalized Recommendations
- Targeted Marketing



C. WEB STRUCTURE MINING

Definition

- Analyze hyperlink structure of the web.

Web Graph

- Nodes = Web Pages
- Edges = Hyperlinks



Important Terms

In-Degree

- Number of incoming links.

Out-Degree

- Number of outgoing links.

Directed Path

- Sequence of directed links.

Shortest Path

- Minimum links needed to reach destination.

Diameter

- Longest shortest path in entire web graph.

Hubs & Authorities

Hub

- Points to many good pages.
- Directory-like page.

Example:

"Top AI Resources" page.

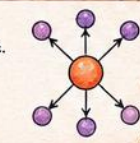


Authority

- Valuable information source.
- Referenced by many hubs.

Example:

Official research website.



Benefits

- Better Search Ranking
- Better Indexing
- Reduced Irrelevant Results



5. E-COMMERCE APPLICATIONS

Customer Analysis

- Predict buying behavior.
- Acquire new customers.
- Retain existing customers.

User Intent Detection

- Determine whether user is:
Buying
Comparing
Browsing

Web Personalization

- Customize website content.
- Based on:
Page Views
Clicks
Searches
Purchases

Product Search & Recommendation

- Analyze search logs.
- Discover product relationships.
- Recommend related products.

WEB MINING

6. SOCIAL MEDIA MINING

Definition

- Extract useful knowledge from social media platforms.

Data Sources

- Facebook
- Twitter/X
- Instagram
- Blogs
- Reviews
- Comments



Social Network Analysis (SNA)

- Nodes = People/Groups
- Links = Relationships

Goal:

Understand social interactions.

Graph Mining

- Works on structured networks.
- Detect communities.
- Find influential users.
- Detect anomalies.



Text Mining

- Works on textual content.
- Analyze:
Comments
Messages
Reviews
Blogs



Applications:

- Sentiment Analysis
- Spam Detection
- Document Classification
- Trend Detection



Data Mining vs Web Mining

Data Mining

- Structured data.
- Relational databases.
- Rows, columns, keys.
- Example:
Student database.

Web Mining

- Semi-structured data.
- Unstructured data.
- HTML pages, text, images, audio and videos.
- Example:
Google Search.

Types of Web Data

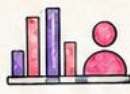
- Web Content
- Web Structure
- Web Usage
- Supplemental Data



Three Categories of Web Mining



Web Content Mining



Web Usage Mining



Web Structure Mining

Web Graph

- Nodes = Web Pages
- Edges = Hyperlinks



Important Terms

- In-Degree - Number of incoming links.
- Out-Degree - Number of outgoing links.
- Directed Path - Sequence of directed links.
- Shortest Path - Minimum links needed to reach destination.
- Diameter - Longest shortest path in entire web graph.

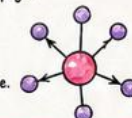
Hubs & Authorities

Hub

- Points to many good pages.
- Directory-like page.

Example:

"Top AI Resources" page.



Authority

- Valuable information source.
- Referenced by many hubs.

Example:

Official research website.



Benefits

- Better Search Ranking
- Better Indexing
- Reduced Irrelevant Results



MOST IMPORTANT EXAM TOPICS

1. Data Mining vs Web Mining
2. Types of Web Data
3. Three Categories of Web Mining
4. Classification vs Clustering
5. Three Phases of Web Usage Mining
6. Association Rules
7. In-Degree vs Out-Degree
8. Hub vs Authority
9. Web Graph
10. Social Network Analysis (SNA)
11. Graph Mining vs Text Mining
12. Web Personalization
13. Product Recommendation
14. Dynamic Nature of Web
15. Challenges of Web Mining

1. WEB MINING FUNDAMENTALS

Definition

- Application of data mining techniques on web data.
- Purpose: Discover useful knowledge, patterns and relationships.



Data Mining vs Web Mining

Data Mining	Web Mining
• Structured data. • Relational databases. • Rows, Columns, Keys.	• Semi-structured data. • Unstructured data. • HTML pages. • Images, videos, text. • Hyperlinks.
Example: Student database Bank database 	Example: Google Search News Websites 

Web Characteristics



- Huge & Wide Coverage**
- Massive amount of information.
 - Almost every topic available.



- Semi-Structured**
- HTML-based documents.



- Highly Linked**
- Connected through hyperlinks.



- Noisy**
- Ads
 - Menus
 - Copyright notices



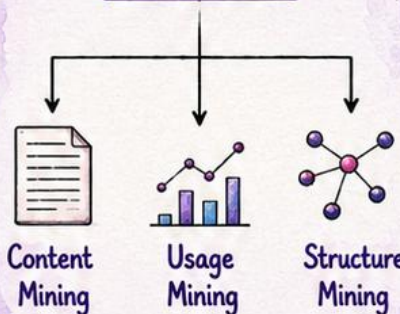
- Redundant**
- Same information repeated on multiple websites.



- Dynamic**
- New pages continuously created.
 - Existing pages updated frequently.

2. THREE CATEGORIES OF WEB MINING

WEB MINING



A. WEB CONTENT MINING

Definition

- Extract information from the actual content of web pages.



Data Sources

- Text
- Images
- Videos
- Audio
- Graphs



Goal

- Determine relevance of content to search queries.



Techniques

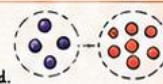
Classification

- Place documents into predefined categories.

Example:
Science
Politics
Sports

Clustering

- Group similar documents.
- No predefined labels needed.



Summarization

- Reduce document length.
- Keep main ideas intact.



Visualization

- Graphical representation of information.

Examples:
Word Clouds
Scatter Plots
Heat Maps



Benefits

- Better search results.
- Easier information retrieval.
- Topic discovery.



Applications

- Search Engines
- Digital Libraries
- Trend Petals



WEB MINING & SOCIAL NETWORK MINING

6. SOCIAL MEDIA MINING

Definition

- Extract useful knowledge from social media platforms.

Data Sources

- Facebook
- Twitter/X
- Instagram
- Blogs
- Reviews
- Comments



Social Network Analysis (SNA)

- Nodes = People/Groups
- Links = Relationships

Goal:

Understand social interactions.



Graph Mining

- Works on structured networks.
- Detect communities.
- Find influential users.
- Detect anomalies.



Text Mining

- Works on textual content.
- Analyze:
 - Comments
 - Messages
 - Reviews
 - Blogs



Applications:

- Sentiment Analysis
- Spam Detection
- Document Classification
- Trend Detection



B. WEB USAGE MINING

Definition

- Discover user browsing patterns.
- Understand user behavior.



Main Data Source

- Server Log Files (Web Logs)



Information Stored

- Clicks
- Page Views
- Form Submissions
- User Sessions



Goals

- Predict future behavior.
- Improve websites.
- Support personalization.



Three Core Phases

1. Data Preprocessing

- Data Cleaning
- Data Integration
- Data Transformation

Result:
Raw logs → Structured logs

2. Pattern Discovery

- Apply mining algorithms.
- Find hidden patterns.

Techniques:

- Classification
- Clustering
- Association Rules



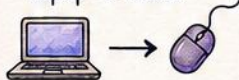
3. Pattern Analysis

- Evaluate patterns.
- Predict future behavior.



Example

- User visits Laptop page.
- Then visits Mouse page.
- Pattern discovered:
Laptop → Mouse



Applications

- Customer Analysis
- Personalized Websites
- Marketing
- Product Recommendation

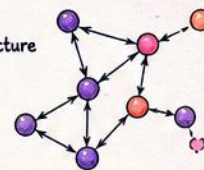


WEB MINING & SOCIAL NETWORK MINING

C. WEB STRUCTURE MINING

Definition

- Analyze hyperlink structure of the web.



Web Graph Model

- Nodes = Web Pages
- Edges = Hyperlinks



Objectives

- Improve search quality.
- Reduce irrelevant results.
- Better indexing.

Important Terms

In-Degree

- Number of incoming links.

Formula:

Links pointing TO page P



Out-Degree

- Number of outgoing links.

Formula:

Links leaving page P



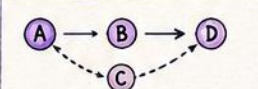
Directed Path

- Sequence of directed links.



Shortest Path

- Minimum links required to reach destination.



Diameter

- Longest shortest path in web graph.



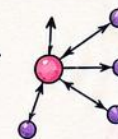
Hubs & Authorities

Good Hub

- Points to many good authorities.

Example:

Resource directory page.

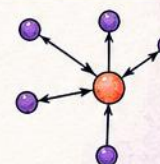


Good Authority

- Receives links from many good hubs.

Example:

Official Wikipedia page
Research institution page



Importance

- Search Engine Ranking
- Link Analysis
- Website Importance Measurement



3. SOCIAL NETWORK MINING

Definition

- Extract patterns and trends from social media data.



Social Network Analysis (SNA)

- Mathematical and visual mapping of relationships.

Purpose:

Understand interactions between people, groups and organizations.



Components

Nodes

- People
- Groups
- Organizations



Links

- Friendships
- Communication
- Relationships
- Information Flow



Goals of SNA

- Find influential users.
- Detect communities.
- Understand information spread.
- Analyze social behavior.



Core Techniques



Graph Mining

- Works on network structure.
- Extracts patterns from graphs.

Applications:

- Community Detection
- Influencer Detection
- Fraud Detection
- Anomaly Detection



Text Mining

- Works on textual content.

Data Sources:

- Blogs
- Reviews
- Tweets
- Comments
- Messages



Applications:

- Sentiment Analysis
- Spam Detection
- Trend Analysis
- Document Classification



Examples

- Twitter/X trend analysis
- Facebook friendship analysis
- YouTube comment analysis



4. E-COMMERCE APPLICATIONS

Customer Analysis

Goal:

Understand customer behavior.

Questions Answered:

- Will customer buy?
- What product will be bought?
- Is user browsing or shopping?



Web Personalization

Definition:

Automatically customize website according to user behavior.

Uses:

- Page Views
- Click History
- Search History
- Form Entries



Examples:

- Amazon recommendations
- Netflix homepage personalization

Product Search

- Analyze search logs.
- Discover relationships between products and queries.

Search...



Product Recommendation

- Recommend related products.
- Based on previous searches.
- Based on previous purchases.

Example:

Search Laptop
→ Recommend Mouse, Bag, Keyboard



WEB MINING & SOCIAL NETWORK MINING

1. WHAT IS WEB MINING?

Definition

- Web Mining = Applying Data Mining techniques on Web data.

Purpose:

Extract useful information, hidden patterns and knowledge from the World Wide Web.

Mining Analogy

- Gold Mining → Extract gold from earth.
- Web Mining → Extract useful knowledge from web pages and web data.

Sources Used in Web Mining

- Web Documents
- Hyperlinks
- Web Services
- Server Logs

Goal

- Discover patterns.
- Understand users.
- Improve search results.
- Support business decisions.

2. TRADITIONAL DATA MINING VS WEB MINING (VERY IMPORTANT EXAM QUESTION)

Traditional Data Mining

- Structured Data
- Relational Databases
- Tables
- Rows
- Columns
- Keys

Characteristics:

- Clean
- Organized
- Mostly Static

Example:

Student Database
Bank Database

Web Mining

- Semi-Structured Data
- Unstructured Data
- HTML Documents
- Text
- Images
- Videos
- Hyperlinks

Characteristics:

- Dynamic
- Noisy
- Redundant
- Linked

Example:

Web Pages
Blogs
Social Media

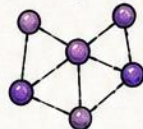
WEB MINING TECHNIQUES (LECTURE 6)

★ QUICK COMPARISON TABLE ★

Feature	Traditional DM	Web Mining
Data Type	Structured	Semi/Unstructured
Storage	Tables	HTML Pages
Organization	Well Organized	Messy
Nature	Static	Dynamic
Quality	Clean	Noisy
Duplication	Low	High (Redundant)
Connectivity	Independent	Hyperlinked
Examples	Databases	Websites

3. THREE MAIN CLASSES OF WEB MINING

WEB MINING



Structure Mining



Content Mining



Usage Mining

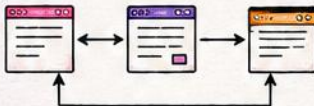
A. WEB STRUCTURE MINING

Definition

- Analyze structure of hyperlinks.

Web Representation

- Nodes = Web Pages
- Edges = Hyperlinks



Purpose

- Understand page relationships.
- Improve search engines.

Example

- Google analyzing links between websites.



B. WEB CONTENT MINING

Definition

- Extract information from page contents.

Content Includes

- Text
- Images
- Videos
- Audio



Main Techniques

Document Classification

Definition:
Assign documents to predefined categories.

Example:

- Science
- Politics
- Sports



Document Clustering

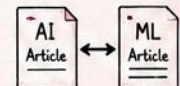
Definition:
Group similar documents automatically.
No predefined labels needed.



Association

Definition:
Discover relationships among content items.

Example:
Users reading AI article also read ML article.



Goal

- Improve relevance of search results.



C. WEB USAGE MINING

Definition

- Analyze user browsing behavior.



Purpose

- Understand users.
- Discover browsing patterns.
- Predict future actions.



Data Source

- Server Logs
- Browser Logs
- Clickstreams



Example

- Which pages users visit most?
- Which products are viewed together?



4. WEB USAGE MINING LOGS (MOST IMPORTANT TOPIC)

Web Usage Mining depends on **LOG FILES**.

A **Log** = Record of user activity stored by server.

There are **FOUR** important logs.

1 TRANSFER / ACCESS LOG

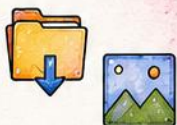
Definition

- Records every request made to the server.



Stores

- Page visits
- File downloads
- Image requests
- URL access



Purpose

- Track website traffic.



Example

User opens:

www.website.com/home

Entry recorded in Access Log.



2 AGENT LOG

Definition

- Records User Agent information.



User Agent Means

- Browser being used.
- Operating System being used.



Examples

Browser:

- Chrome
- Firefox
- Safari
- Edge



Operating System:

- Windows
- Linux
- Android
- iOS



Purpose

- Understand visitor technology.



3 REFERRER LOG

Definition

- Records previous webpage visited by user.



Answers Question

"Where did user come from?"



Example

User clicks:
Google Search
↓
Your Website

Referrer:
google.com



Purpose

- Traffic source analysis.
- Marketing analysis.



4 ERROR LOG

Definition

- Records server and website errors.



Examples

404

404 Error
Page Not Found

500

500 Error
Internal Server Error



Database Failure



Server Crash

Purpose

- Troubleshooting.
- Website maintenance.
- Error monitoring.

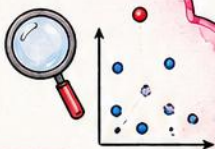


WEB MINING TECHNIQUES

1. INTRODUCTION

Definition

- An outlier is a data object that deviates significantly from the majority of the dataset.
- Appears abnormal, unusual, rare or suspicious.



Other Names

- Anomaly
- Exception
- Deviation



Goal of Outlier Detection

- Identify abnormal patterns.
- Detect unusual events.
- Discover rare but important cases.



Real-Life Examples

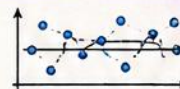
- Credit Card Fraud
- Network Intrusion Detection
- Medical Diagnosis
- Insurance Fraud
- Equipment Failure Detection



2. OUTLIERS VS NOISE

Noise

- Random error.
- Small variation.
- Usually ignored or removed.



Example:

Customer buys one extra coffee.



Outlier

- Important abnormal event.
- Requires investigation.
- May indicate fraud or attack.



Example:

Customer suddenly spends \$50,000 in another country.



Exam Point:

Noise = Error

Outlier = Interesting Information

3. TYPES OF OUTLIERS (VERY IMPORTANT)

1 Global Outlier (Point Anomaly)

Definition:

- A single point that is far away from all other data points.

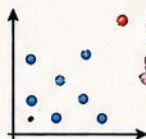
Characteristics:

- Clearly abnormal.
- Easy to identify.

Example:

Heights:

5.2, 5.4, 5.5, 5.7, 5.8, 7.5
→ 7.5 is Global Outlier



2 Contextual Outlier (Conditional Outlier)

Definition:

- Outlier only in a specific context.

Context Examples:

- Time
- Location
- Season

Example:

Temperature = 28°C

Summer → Normal

Winter → Outlier

Same value, different context.



3 Collective Outlier

Definition:

- Individual objects may be normal.
- Entire group becomes abnormal.

Example:

One delayed order → Normal

100 delayed orders together

→ Collective Outlier



Indicates:

- Supply chain failure
- System malfunction



★ Memory Trick:

- Global = One abnormal point
- Contextual = Abnormal in context
- Collective = Abnormal group

OUTLIER (ANOMALY) DETECTION

4. APPLICATIONS OF OUTLIER DETECTION

Fraud Detection

- Credit card fraud
- Insurance fraud



Cybersecurity

- Intrusion detection
- Network attack detection



Medical Analysis

- Disease detection
- Abnormal patient behavior



Manufacturing

- Defective products
- Machine failures



Financial Analysis

- Suspicious transactions
- Market manipulation detection



5. CHALLENGES IN OUTLIER DETECTION

Blurry Boundary Problem

- Hard to define exact line between normal and abnormal.



Application Dependency

- Outlier definition changes from one domain to another.

Example:

Medical systems need strict anomaly detection.



Noise Handling

- Noise may look like outliers.
- Outliers may hide inside noise.



Data Quality Issues

- Missing values.
- Incorrect data.



Explainability

- System should explain WHY object was marked as outlier.



6. LEARNING APPROACHES

1. Supervised Methods

Definition:

- Training data already labeled.

Labels:

Normal

Outlier

Advantages:

- High accuracy.

Limitation:

- Hard to obtain enough outlier examples.



3. Semi-Supervised Methods

Definition:

- Few labeled examples available.

Uses:

- Small labeled data
- + Large unlabeled data

Goal:

- Learn normal behavior.

2. Unsupervised Methods

Definition:

- No labels available.

Assumption:

- Normal data forms clusters.
- Isolated points are outliers.

Advantages:

- Most widely used.

Limitation:

- May produce false alarms.



★ Memory Trick:

Supervised = Fully labeled

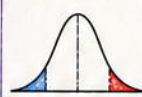
Semi-supervised = Partially labeled

Unsupervised = No labels

7. OUTLIER DETECTION TECHNIQUES

OUTLIER DETECTION

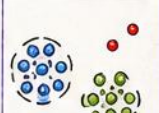
Statistical (Model-Based)



Proximity-Based



Clustering-Based



OUTLIER (ANOMALY) DETECTION

A. STATISTICAL APPROACHES (MODEL-BASED)

Idea

- Assume normal data follows a statistical distribution.
- Objects violating the model become outliers.

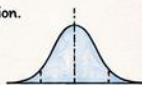


Parametric Methods

Definition:

- Assume specific distribution.

Example:

Normal Distribution
Bell Curve

Important Techniques:

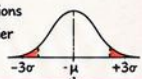
Boxplot

- Values outside whiskers are outliers.



3-Sigma Rule

- Mean $\pm$ 3 Standard Deviations
- Outside this range = Outlier



Non-Parametric Methods

Definition:

- No predefined distribution.

Example:

Histogram

Rule:

- Rare or empty bins indicate potential outliers.



B. PROXIMITY-BASED APPROACHES

Idea

- Examine nearby neighbors.

Distance-Based Methods

Definition:

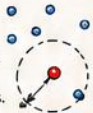
- Count neighbors within radius.

Rule:

- Few neighbors $\rightarrow$ Outlier

Characteristics:

- Isolated objects are suspicious.



Density-Based Methods

Definition:

- Compare local density.

Rule:

- Lower density than neighbors $\rightarrow$ Outlier

Example:

- Object in empty region while neighbors are densely packed.



C. CLUSTERING-BASED APPROACHES

Idea

- Group similar objects into clusters.

Assumption

- Normal data belongs to large dense clusters.

Outliers Are

- Objects in very small clusters

OR

- Objects not belonging to any cluster

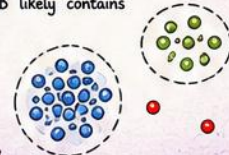


Example

Cluster A = 1000 points

Cluster B = 2 points

- $\rightarrow$ Cluster B likely contains outliers



★ COMPARISON OF TECHNIQUES ★

Statistical Approach

- Uses probability distributions.
- Example: 3-Sigma Rule, Boxplot.



Distance-Based Approach

- Uses neighbor distance.
- Isolated points are outliers.



Density-Based Approach

- Uses local density.
- Sparse region = outlier.



Clustering-Based Approach

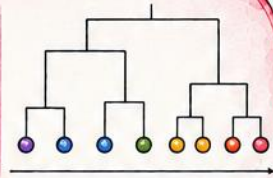
- Uses clusters.
- Small clusters = outliers.



1. HIERARCHICAL CLUSTERING

Definition

- Creates a hierarchy (tree) of clusters.
- Result represented by: **DENDROGRAM** (Tree Structure)



Advantages

- Easy visualization.
- Good for summarization.
- Reveals hidden structures.
- Shows relationships among clusters.

Types

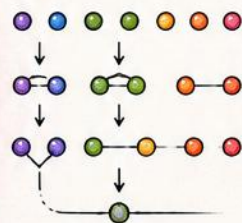
Agglomerative (Bottom-Up)

Process:

- Step 1 → Each object = separate cluster
- Step 2 → Merge closest clusters
- Step 3 → Continue merging
- Step 4 → One root cluster remains

Example:

AGNES
(Agglomerative Nesting)



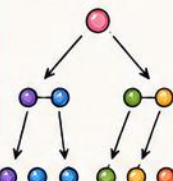
Divisive (Top-Down)

Process:

- Step 1 → One big cluster
- Step 2 → Split into subclusters
- Step 3 → Continue splitting
- Step 4 → Small clusters formed

Example:

DIANA
(Divisive Analysis)



★ Memory Trick

AGNES = Merge
DIANA = Divide

2. LINKAGE METHODS (Distance Between Clusters)

Single Linkage (Minimum Distance)

Definition:

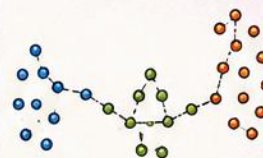
- Distance between closest pair of objects.

Characteristics:

- Forms chain-like clusters.
- Sensitive to noise.

Example:

Minimum Spanning Tree concept.



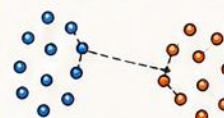
Complete Linkage (Maximum Distance)

Definition:

- Distance between farthest pair of objects.

Characteristics:

- Compact clusters.
- Small cluster diameter.



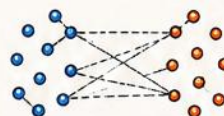
Average Linkage

Definition:

- Average distance between all objects.

Characteristics:

- Balanced approach.



Mean Linkage

- Uses mean distances.
- Less sensitive to outliers.

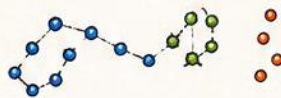


CLUSTERING METHODS

★ IMPORTANT COMPARISON ★

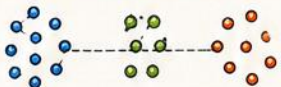
Single Linkage

- Closest points
- Chain effect
- Sensitive to noise



Complete Linkage

- Farthest points
- Compact clusters



Average Linkage

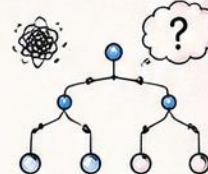
- Average distance
- Balanced performance



3. ADVANCED HIERARCHICAL METHODS

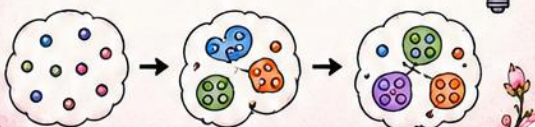
Basic Hierarchical Problems:

- Cannot undo merge decisions
- Cannot undo split decisions
- Poor scalability for huge datasets



Solution:

Advanced Multiphase Methods



A. BIRCH

(Balanced Iterative Reducing and Clustering using Hierarchies)

Purpose

- Designed for very large datasets.
- Mainly numeric data.

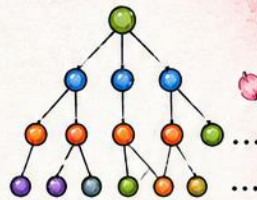
Main Idea

- Summarize clusters using Clustering Features (CF).

CF Vector

$$CF = \langle n, LS, SS \rangle$$

n = Number of points
LS = Linear Sum
SS = Square Sum



Data Structure

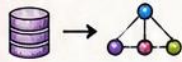
CF Tree

Compact representation of huge datasets.

Phases

Phase 1:

- Scan data.
- Build CF Tree.



Phase 2:

- Apply clustering algorithm on leaf nodes.



Time Complexity

$O(n)$

Advantages

- Very fast.
- Scalable.
- Memory efficient.

Limitation

- Poor for non-spherical clusters.



B. CHAMELEON

Main Idea

- Uses k-Nearest Neighbor Graph.



Process

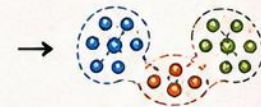
Step 1:

Create sparse graph.



Step 2:

Create subclusters.



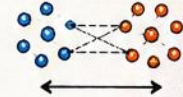
Step 3:

Merge similar subclusters.



Uses Two Measures

- Relative Interconnectivity
- Relative Closeness



Advantage

- Finds complex cluster shapes.

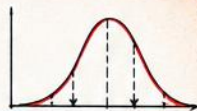


CLUSTERING METHODS

C. PROBABILISTIC HIERARCHICAL CLUSTERING

Main Idea

- Uses probability models.
- Uses generative models.



Example

Gaussian Distribution



Goal

Maximize likelihood of data fitting model.



Advantages

- Handles uncertainty.
- Handles missing values.



Application

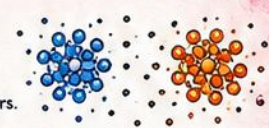
- Incomplete datasets.



4. DENSITY-BASED CLUSTERING

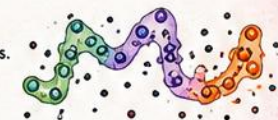
Definition

- Clusters are dense regions.
- Sparse regions separate clusters.



Major Advantage

- Finds arbitrary-shaped clusters.
- Handles noise well.



Main Algorithms

- DBSCAN
- OPTICS
- DENCLUE



A. DBSCAN

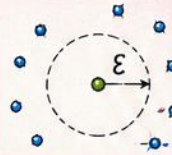
(Density-Based Spatial Clustering of Applications with Noise)

Parameters

- ϵ (Epsilon)
- Radius

MinPts

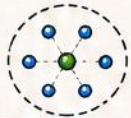
- Minimum neighbors required



Core Object

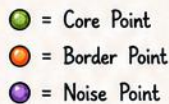
Definition:

Object having at least MinPts neighbors within ϵ .



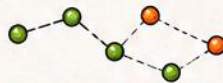
Concepts

- Core Point
- Border Point
- Noise Point



Connectivity

- Density Reachable
- Density Connected



Advantages

- Finds arbitrary shapes.
- Handles noise.



Time Complexity

With Index: $O(n \log n)$

Without Index: $O(n^2)$



Limitation

- Sensitive to parameter selection.



B. OPTICS

(Ordering Points To Identify Clustering Structure)

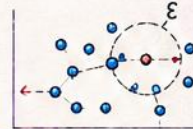
Purpose

- Improvement over DBSCAN.



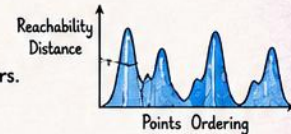
Problem Solved

- DBSCAN sensitive to global parameters.



Output

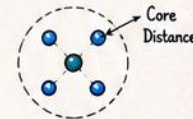
- Cluster Ordering
- Instead of strict clusters.



Important Terms

Core Distance

- Distance needed for point to become core object.



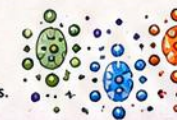
Reachability Distance

- Distance needed to make point reachable.



Advantage

- Detects clusters with varying densities.



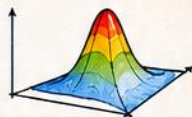
CLUSTERING METHODS

C. DENCLUE

(Density-based CLUstering)

Main Idea

- Uses mathematical density functions.

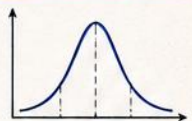


Method

Kernel Density Estimation

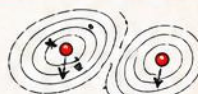
Example:

Gaussian Kernel



Cluster Centers

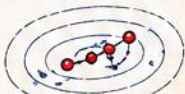
Density Attractors
(Local Maximum Density Points)



Process

Hill Climbing Procedure

→ Move toward highest density.



Advantage

- Strong mathematical foundation.



5. GRID-BASED CLUSTERING

Definition

- Divides data space into grid cells.



Main Idea

Space Driven

Instead of:

Data Driven



Advantage

- Very fast processing.
- Works well with huge datasets.



Main Algorithms

- STING
- CLIQUE

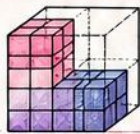


A. STING

(Statistical Information Grid)

Main Idea

- Divide space into hierarchical rectangular cells.



Information Stored

Each cell stores:

- Count
- Mean
- Minimum
- Maximum
- Standard Deviation

#	$\bar{x}$
Min	Max
σ	

Advantages

- Fast query processing.
- Efficient clustering.



Complexity

Clustering: $O(n)$

Querying: $O(g)$

g = Number of grid cells



Suitable For

- Spatial databases.



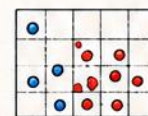
B. CLIQUE

Full Form

Clustering In QUEST

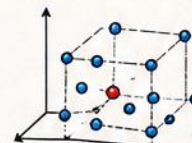
Type

- Grid-Based
- Density-Based



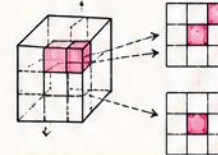
Purpose

- High-dimensional data.



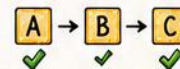
Main Idea

- Dense high-dimensional cell exists only if all lower-dimensional projections are dense.



Uses

Apriori Principle



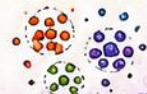
Additional Principle

MDL
(Minimum Description Length)



Advantage

- Discovers clusters in high-dimensional subspaces.



CLUSTERING METHODS

★ ALGORITHM COMPARISON ★

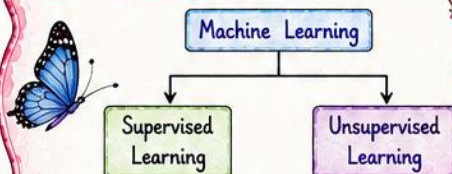
AGNES		- Hierarchical - Bottom-Up - Merge clusters
DIANA		- Hierarchical - Top-Down - Split clusters
BIRCH		- Hierarchical - Uses CF Tree $O(n)$
CHAMELEON		- k-NN Graph - Relative Interconnectivity
DBSCAN		- Density-Based - Uses ϵ and MinPts
OPTICS		- Improved DBSCAN - Uses Core Distance
DENCLUE		- Kernel Density - Density Attractors
STING		- Grid-Based - Hierarchical Cells
CLIQUE		- Grid + Density - High-Dimensional Data

MACHINE LEARNING OVERVIEW

Definition

- Field of Artificial Intelligence that enables computers to learn from data without being explicitly programmed.

Main Categories



Purpose

- Learn patterns from data.
- Make predictions.
- Discover hidden relationships.

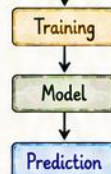
3. SUPERVISED LEARNING

Definition

- Learning from labeled data.
- Training data already contains correct answers (labels).
- Acts like a teacher guiding a student.

Main Idea

Input Data + Correct Answers



Workflow

- Step 1:
Collect labeled data
- ↓
- Step 2:
Train model
- ↓
- Step 3:
Test model
- ↓
- Step 4:
Generate predictions

Goal

- Predict unknown outputs.

4. TYPES OF SUPERVISED LEARNING

A. REGRESSION

Definition:

- Predicts continuous numeric values.

Target Variable:

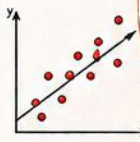
- Real-valued (continuous)

Examples:

- House Price Prediction
- Temperature Prediction
- Stock Price Prediction
- Sea Ice Extent Prediction

Output Examples:

- 45.6
- 100.2
- 250000



B. CLASSIFICATION

Definition:

- Predicts categories/classes.

Target Variable:

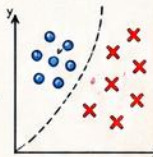
- Categorical

Examples:

- Spam / Not Spam
- Pass / Fail
- Disease / Healthy
- Benign / Malignant Tumor

Output Examples:

- Yes / No
- 0 / 1
- Class A / B / C



MACHINE LEARNING OVERVIEW

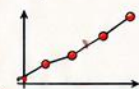
REGRESSION VS CLASSIFICATION

Regression

- Numeric Output
- Continuous Values

Examples:

- House Price
- Temperature

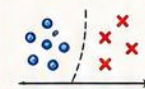


Classification

- Categorical Output
- Class Labels

Examples:

- Spam Detection
- Cancer Classification



5. SUPERVISED LEARNING ALGORITHMS



Linear Regression

- Predicts continuous values.

Example:

House price prediction.



Logistic Regression

- Classification algorithm.

Example:

Disease prediction.

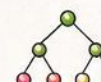


Decision Trees

- Tree-like decision structure.

Example:

Loan approval.



Support Vector Machine (SVM)

- Separates classes using optimal boundaries.

Example:

Image classification.



Neural Networks

- Brain-inspired model.
- Used in Deep Learning.
- Handles complex patterns.



6. ADVANTAGES OF SUPERVISED LEARNING

- High Accuracy
- Easy Evaluation
- Clear Learning Process
- Strong Prediction Capability
- Works Well When Labels Exist



7. DISADVANTAGES OF SUPERVISED LEARNING

- Requires Labeled Data
- Labeling Is Expensive
- Data Preparation Takes Time
- Large Training Data Needed
- Human Effort Required

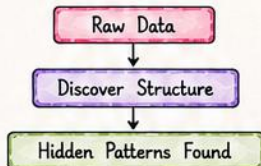


8. UNSUPERVISED LEARNING

Definition

- Learning from unlabeled data.
- No teacher or supervision.
- System discovers patterns itself.

Main Idea



Goal

- Find hidden relationships.
- Group similar objects.
- Compress data.
- Discover structures.

Characteristics

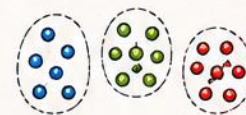
- No labels.
- Exploratory analysis.
- Pattern discovery.



9. TYPES OF UNSUPERVISED LEARNING

Clustering

- Group similar objects.

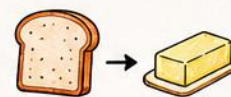


Association

- Find relationships between items.

Example:

Bread → Butter

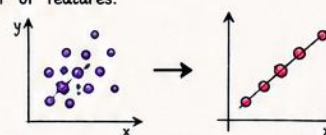


Dimensionality Reduction

- Reduce number of features.

Example:

PCA



10. ADVANTAGES OF UNSUPERVISED LEARNING

- ✓ No Labeling Needed
- ✓ Saves Time
- ✓ Discovers Hidden Patterns
- ✓ Useful for Exploration
- ✓ Handles Large Data

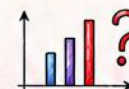


MACHINE LEARNING OVERVIEW



11. DISADVANTAGES OF UNSUPERVISED LEARNING

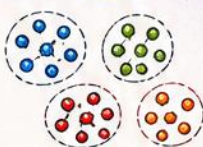
- ✗ Lower Accuracy
- ✗ Hard to Evaluate
- ✗ Results May Be Ambiguous
- ✗ No Ground Truth Available
- ✗ Interpretation Can Be Difficult



12. CLUSTERING (DEEP DIVE)

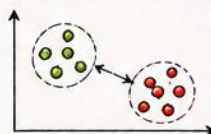
Definition

- Most popular unsupervised learning technique.
- Groups similar objects together.
- Separates dissimilar objects.



Objective

Same Cluster
→ High Similarity
Different Clusters
→ Low Similarity



Nature

- Exploratory Data Analysis.
- Pattern Discovery.
- No Labels required.



Goal

- Reveal hidden structures.



13. APPLICATIONS OF CLUSTERING

Biology

- Species classification.
- Genetic analysis.



Medical Imaging

- Tissue identification.
- Disease detection.



Market Research

- Customer segmentation.
- Consumer behavior analysis.



Recommender Systems

- Amazon recommendations.
- Netflix movie suggestions.



Business Intelligence

- Customer profiling.
- Marketing campaigns.



14. TYPES OF CLUSTERING ALGORITHMS

CLUSTERING

Partition-Based

Hierarchical

Density-Based

A. PARTITION-BASED CLUSTERING

Definition

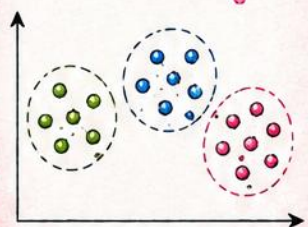
- Divides data into k clusters.

Characteristics

- Fast
- Efficient

Example

- K-Means



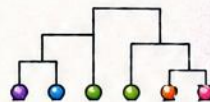
B. HIERARCHICAL CLUSTERING

Definition

- Produces tree-like clusters.

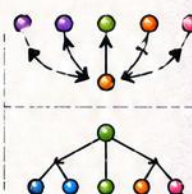
Output

- Dendrogram



Types

- Agglomerative (Bottom-Up)
- Divisive (Top-Down)



Advantage

- Easy visualization.



C. DENSITY-BASED CLUSTERING

Definition

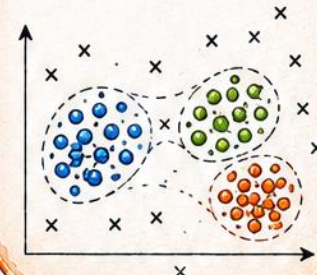
- Forms clusters from dense regions.

Characteristics

- Finds arbitrary shapes.
- Handles noise well.

Example

- DBSCAN



MACHINE LEARNING OVERVIEW

★ SUPERVISED VS UNSUPERVISED ★

Supervised Learning

- ✓ Labeled Data
- ✓ Teacher Available
- ✓ Predict Output
- ✓ Higher Accuracy
- ✓ Examples:
Regression
Classification



VS

Unsupervised Learning

- ✓ Unlabeled Data
- ✓ No Teacher
- ✓ Discover Patterns
- ✓ Exploratory Analysis
- ✓ Examples:
Clustering
Association
Dimensionality Reduction

